

Yale School of Management - MGT404 Introduction to Microeconomics
Practice Problems, Lectures 1-10 (the end of Price Discrimination)

Page 1-8: questions. Pages 9-13 solutions. **Print Pages 1-8 to work through**, only check solutions afterwards.

1. *United Airlines*. United Airlines operates a non-stop route between New York City and Paris. There are two types of travelers: business and tourists. The monthly demand for round-trip flights for these two groups are given by:

$$P_B = 3,000 - \frac{1}{5}Q_B$$

$$P_T = 1,000 - \frac{1}{5}Q_T$$

where B and T denote business and tourist, respectively. Quantities are the total number of round-trip tickets sold per month and prices are on a per-ticket basis.

- a) How many tickets would United sell in a month if it priced this route at \$800 per round-trip ticket?

No. of Tickets: _____

- b) What percent of travelers would be leisure travelers?

Percent leisure: _____

- c) The government is considering imposing a security surcharge on all international tickets. What is the smallest security surcharge (which would be added to the \$800 airfare) that would cause leisure travelers to stop traveling?

Surcharge: _____

- d) Now, ignore the security surcharge. In light of ongoing structural changes occurring in Europe due to Britain's exit from the European Union, United Airlines is evaluating the impact on business travel between New York and Paris. United has estimated that the relocation of some firms from London to Paris, will result in an increase in the demand for business travel between New York and Paris, with the resulting monthly demand for business travelers being:

$$P_B = 4,000 - \frac{1}{5}Q_B$$

How much is this growth in demand going to increase (in percentage terms) the total flights per-month demanded under United's original pricing of \$800 per flight?

% increase = _____

e) If they keep the price fixed, would their customer mix (business versus tourist) shift towards more business travelers or towards more tourists?

- ☐ towards more business travelers
- ☐ towards more tourists
- ☐ stay the same

2. *Philadelphia Soda*. The market for two-liter bottles of soda in Philadelphia is given by the following demand:

$$P_D = 10 - 2Q_D$$

and by the following supply curve:

$$P_S = \frac{1}{2}Q_S$$

where P_D and P_S are the per-bottle prices paid by consumers and received by producers, respectively, and Q is the quantity of bottles in millions per month.

a) What is the equilibrium price?

Price: _____

b) What is the equilibrium quantity?

Quantity: _____

c) Effective this past January, Philadelphia imposed a tax on sweetened beverages. The tax is 1.5 cents-per-ounce, which is roughly a \$1 tax per two-liter bottle. Determine the equilibrium price paid by consumers with a \$1 tax per bottle.

Consumer Price: _____

d) How much revenue does the tax raise?

Revenue (millions): _____

e) Which side of the market “pays” more of the tax burden?

- ☐ Consumers
- ☐ Producers
- ☐ Even split

3. *Trade.* The demand for pineapples in Belize is given by:

$$Q_D = 20 - P$$

And the supply there is given by

$$Q_S = 4P - 5$$

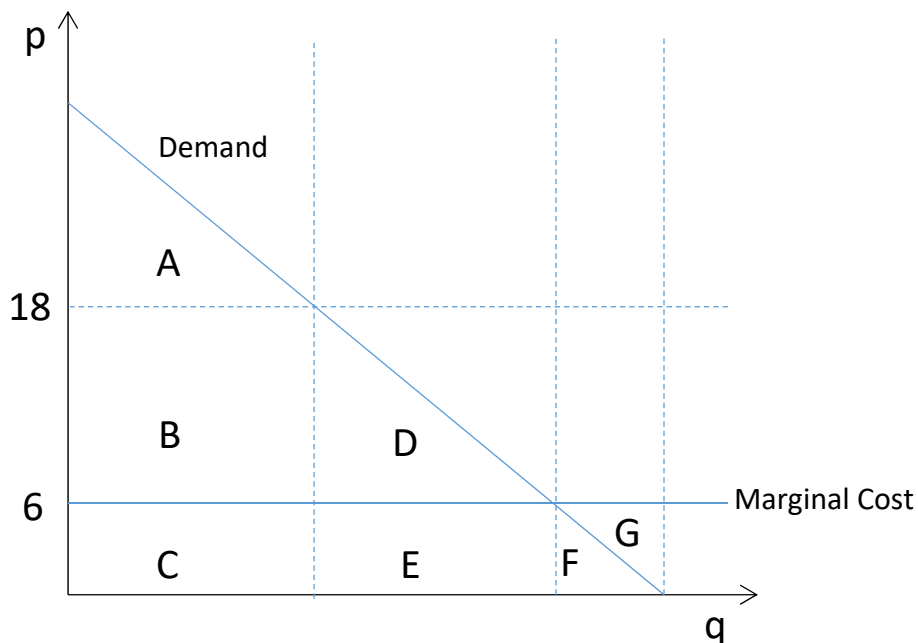
Assume Belize is “small” in that it cannot affect the world price of pineapples.

1. What are the autarky (no trade) price and quantity of pineapples in Belize?
2. Suppose free trade prevails and Belize exports 25 pineapples. What must the world price of pineapples be? (Hint: if you’re unsure how to begin, draw the graph. Recall that when we plot these functions, we write them as $P = \dots$)
3. Now Belize places an export subsidy of 5 on each unit of pineapples exported. What are exports now?
4. What is the welfare loss or gain associated with the subsidy?

4 *Crossfit*. Crossfit is a popular group workout. All of Crossfit's 1000 target customers in Philadelphia are identical and have a monthly demand given by:

$$p = 30 - q$$

The marginal cost of an additional person in the class is \$6. The demand and marginal cost is depicted on the picture below.



(You are not required to show supporting work for this question.)

(a) (5 points). Cross-fit determines that their best uniform, per-class price is \$18 per class.

- i. How many times would the customer go? _____
- ii. Total amount each customer pays
(list the letters from the picture above): _____
- iii. Company profit, per customer (list the letters): _____
- iv. Consumer surplus (list the letters): _____

v. Deadweight loss (list the letters): _____

(b) (6 points). Cross-fit decides to price instead with a two-part tariff. Indicate the area on the graph that would correspond to the membership fee and to the usage fee each customer would pay in an optimal two-part tariff, then describe profit and usage:

i. Area for membership fee (list the letters): _____

ii. Area for usage fee (list the letters): _____

iii. Company profit (list the letters): _____

iv. How many times would the customer go? _____

(c) (4 points). Crossfit now does further market research and realizes there is a second type of consumer in the market who would like to use the Crossfit gyms much less frequently than their current customers. If Crossfit wants to target these potential “light users” to become customers in addition to the frequent users, how should they change their membership fee and their per-use fee to maximize profits?

Membership fee:

☐ Increase it ☐ Decrease it ☐ No change

Per-use fee:

☐ Increase it ☐ Decrease it ☐ No change

5. *Payne-Whitney*. Due to the need for MBA students to work off the thousands of calories consumed from motivational beverages, an individual MBA’s demand for Payne-Whitney gym is estimated to be

$$q_m = 220 - 55p$$

where p is the per-visit price and q is the number of total visits per year. The gym’s marginal cost per visit is so small that you can assume it’s 0.

(a) (4 points). How should Payne-Whitney price an unlimited-access annual membership to the gym?

Optimal membership fee: _____

Of course, Payne-Whitney has customers other than MBAs, such as Undergrads. While MBAs' willingness-to-pay for an annual membership is exactly equal to the membership price found in (a), Undergrads' willingness-to-pay is only \$350. The gym decides to sell two different annual membership plans: a regular one and a special one with all the perks (attend at all hours, pre-register for best classes, reserved changing area, etc.). Through a survey, Payne-Whitney finds that MBAs are willing to pay an additional \$40 for the special membership, while Undergrads are willing to pay an additional \$10 for the special membership (so price found in (a) + \$40 for MBAs, and a total of \$360 for the undergrads for the special membership). For a regular membership, Payne-Whitney still incurs 0 marginal cost, but the perks in the special membership plan create a \$15 cost per special member per year.

(b) (6 points). Suppose there are an equal number of MBAs and Undergrads in the market for Payne-Whitney – think of there being two people where one is an MBA and the other is an Undergrad. How should they price the regular membership plan and the special membership plan?

Regular membership plan: $P =$ _____

Special membership plan: $P =$ _____

Payne-Whitney's profit per two people
(one MBA, one Undergrad): _____

(c) (4 points). If for some reason they had to keep offering just one membership plan, which would it be, and at what price?

☐ REGULAR AT _____

☐ SPECIAL AT _____

How much extra profit per two people (one MBA, one Undergrad) do they make by having the two memberships instead of one?

Extra Profit: _____

6. *Short answer questions*

a) A dating app that had been charging \$9.99 per month ran a pricing experiment: they found that if they increased their price by 20%, the number of users subscribing would fall by 12%. At the price of \$9.99, consumer demand for this service could therefore be described as:

- ☐ Unresponsive
- ☐ Elastic
- ☐ Inelastic
- ☐ None of the above

b) After Philadelphia imposed a sweetened-drink tax, sales of bottled water increased by approximately 10%. Select the option below that correctly describes this relationship.

- ☐ The cross-price elasticity of water with respect to soda is **positive** as the two goods are **substitutes**.
- ☐ The cross-price elasticity of water with respect to soda is **negative** as the two goods are **substitutes**.
- ☐ The cross-price elasticity of water with respect to soda is **positive** as the two goods are **complements**.
- ☐ The cross-price elasticity of water with respect to soda is **negative** as the two goods are **complements**.

c) This year, a software firm has determined that to optimally produce products, it should employ four analysts for every one manager. However, due to high competition for talent, the salaries for managers are expected to rise by 10% next year. The optimal ratio of analysts to managers next year will be:

- ☐ Lower than this year
- ☐ The same as this year
- ☐ Higher than this year

d) A bakery determined that its total cost function is to produce q loaves of high-quality bread per week is given by

$$TC(q) = 400 + \sqrt{q}$$

Match the equations and labels below.

<i>Fixed Cost</i>	$\frac{400 + \sqrt{q}}{q}$
<i>Variable Cost</i>	400
<i>Average Cost</i>	$\frac{1}{2\sqrt{q}}$
<i>Marginal Cost</i>	$\sqrt{q}$

- e) If a bakery has determined that it can sell loaves of its high quality bread for \$5, and has a total cost function of:

$$TC(q) = 50 + 0.1q^2$$

How many will it choose to produce in a given week?

SOLUTIONS

Advice: ALWAYS attempt problems on your own, then discuss with learning team, before looking at solutions.

Problem 1: United

At a price of \$800, United would sell:

$$800 = 3000 - \frac{1}{5} Q_b$$

$$\frac{1}{5} Q_b = 2200$$

$$Q_b = 11,000$$

11,000 tickets to business travelers, and

$$800 = 1000 - \frac{1}{5} Q_t$$

$$\frac{1}{5} Q_t = 200$$

$$Q_t = 1000$$

An additional 1000 tickets to tourists, for a total of 12,000.

The percent of leisure would be $1/12 = 8.3\%$, or 8% rounded.

From the demand function, it is clear that no leisure traveler would buy a ticket if the price were above \$1000. That is, $Q_t = 0$ if $P = 1000$. So, a surcharge of \$200 would price out leisure travelers.

Business travel demand now increases: their demand is now

$$800 = 4000 - \frac{1}{5} Q_b$$

$$\frac{1}{5} Q_b = 3200$$

$$Q_b = 16000$$

Tourist demand is unaffected (at 1,000). So, total goes from 12,000 to 17,000, an increase of 41.67% (rounded to 42%).

This clearly shifts the mix toward business.

Problem 2: Philly Soda Tax

Equilibrium price and quantity come from equating supply and demand:

$$10 - 2Q = \frac{1}{2} Q$$

$$10 = \frac{5}{2} Q$$

$$4 = Q$$

At a supply of 4, price is

$$P = 10 - 2(4) = 2$$

So, price is \$2 and quantity is 4M bottles.

The tax implies $P_{tax} = P_s + 1 = \frac{1}{2}Q_s + 1$

The new equilibrium sets supply equal to demand, or

$$10 - 2Q = \frac{1}{2}Q + 1$$

$$9 = \frac{5}{2}Q$$

$$q = 3.6$$

Plugging back in to demand,

$$10 - 2(3.6) = 2.80$$

This is the new equilibrium price.

The tax raises $q \cdot \$1 = 3.6M$ in revenue.

Which side pays the tax?

Using one of the shortcut methods, equilibrium price without tax was \$2, and now producers receive \$1.80 (a \$0.20 drop) while consumers pay \$2.80 (a \$0.80 increase). So, consumers “pay” the tax.

Problem 3, Trade

1. Solve for the equilibrium by setting demand = supply

$$Q_D = Q_S$$

$$20 - P = 4P - 5$$

$$25 = 5P$$

$$P^* = \$5$$

$$Q^* = 15$$

2. Belize is exporting 25 pineapples. This means that the quantity supplied must be the quantity demanded domestically at the world price + 25.

$$Q_S(P^W) = Q_D(P^W) + 25$$

$$4P^W - 5 = 20 - P^W + 25$$

$$5P^W = 50$$

$$P^W = \$10$$

So the world price is \$10. It is easy to now check that $Q_D(10) = 10$, $Q_S(10) = 35$, and so the country must be exporting 25 pineapples.

3. This export subsidy has the effect of increasing the world price to \$15. At that price, we have

$$Q_S(15) = 55$$

$$Q_D(15) = 5$$

Producers would like to supply 55 units at a world price of \$10 + an export subsidy of \$5. Since they only get the export subsidy if they export, any domestic buyers would have to agree to pay \$15 as well (otherwise the producer would export at the world price of \$10 and receive the \$5 export subsidy).

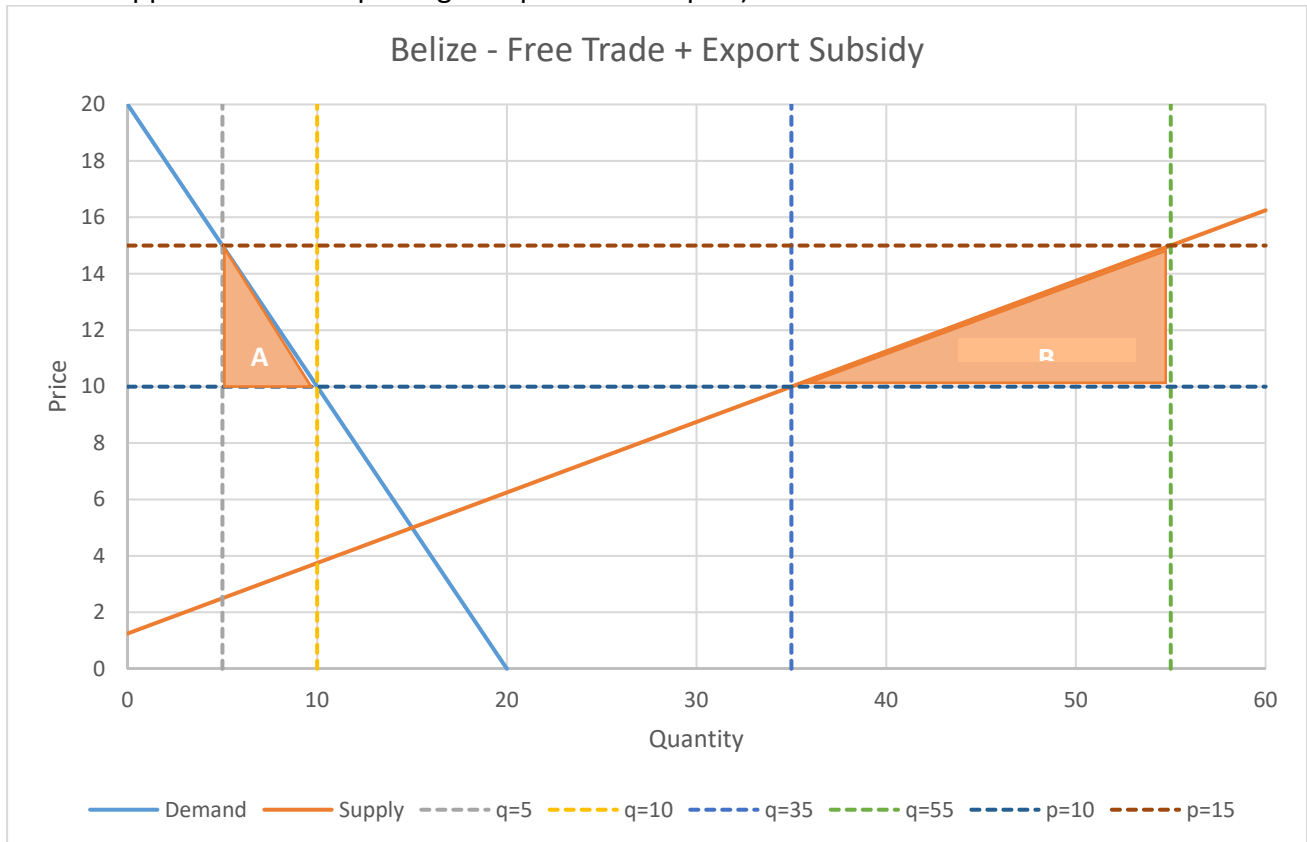
As a result, only 5 units are demanded domestically, and so 50 units are exported.

4. There are two deadweight loss regions created by this subsidy (see diagram on following page). The triangles have a combined area of 62.5, so the welfare loss is 62.5.

$$\frac{1}{2}(5)(5) + \frac{1}{2}(20)(5) = \frac{1}{2}(25) + \frac{1}{2}(100) = 62.5$$

Area A: Before subsidy: consumer surplus. After subsidy: producer surplus instead of consumer surplus, but also government expenditure

Area B: Before subsidy: nothing. After subsidy: government expenditure (note that suppliers are not capturing it as producer surplus).



Problem 4: Crossfit

(a) (5 pt)

How many times? : 12 (1 pt)

Total amount each pays? : B+C (1 pt) Company profit : B (1 pt)

CS: A (1 pt)

DWL: D (1 pt)

(b) (6 pt)

Area for membership fee: A+B+D (2 pt)

Area for usage fee: C+E (2 pt)
Company profit: A+B+D (1 pt)
How many times? : 24 (1 pt)

(c) (4 pt)

Membership fee: Decrease it (2 pt)
Per-use fee: Increase it (2 pt)

If the light users have lower willingness to pay, then they will not join at the current membership fee. They would have to shift away from membership fee towards usage fee.

Question 5: Payne-Whitney

(a)

Optimal membership fee: 440

The optimal membership fee is equal to the consumer surplus when the per-visit price is $MC=0$:

$$CS = 0.5 \times 4 \times 220 = 440$$

(b)

Regular membership plan: 350 (2 pt)
~~Special membership plan: 390 (3 pt)~~
Payne-Whitney's profit: 725 (1 pt)

Optimal to make MBAs buy the special plan, and the Undergrads/MAs the regular plan.
Price the regular membership such that it leaves Undergrads/MAs with zero surplus:

$$P_R = 350$$

Price the special membership plan such that the MBAs are indifferent between the special and the regular plan

$$480 - P_S = 440 - 350$$

$$P_S = 390$$

Payne-Whitney's profit from one MBA and one Undergrad/MA is:

$$\pi = 350 + 390 - 15 = 725$$

(c)

Regular at 350

Extra Profit: 25

If they sell only the regular plan, optimal to price at 350 and earn $350 \times 2 = 700$. If it sells only the special plan, optimal to price at 360 and earn $2 \times (360 - 15) = 690$. So they are better off selling the regular plan at 350.

Payne-Whitney earns additional profit of $725 - 700 = 25$.

Short answer questions:

a. Inelastic. If the demand had fallen by more than 20%: elastic. Since it fell by less, "inelastic".

b. positive, substitutes.

c. as the input price rises, we substitute away, so "higher".

d. $FC = 400$, $VC = q^{1/2}$, $AVC = (400 + q^{1/2})/q$, $MC = 1/(2q^{1/2})$

e. Marginal cost function is $0.2q$, so supply curve is $0.2q$. If the price is \$5,

$$\$5 = 0.2q$$

$$25 = q$$

They will produce 25.